THE UNIVERSITY OF BRITISH COLUMBIA

Dear Dr. Öpik

Please consider our paper, "Traits predict forest phenological responses to photoperiod more than temperature" for publication as a full paper in *New Phytologist*.

Climate change is altering the environmental cues of growth globally, producing shifts in the timing of life history events (phenology), altering resource use and producing novel species interactions. But to date, we have observed considerable variation in communities' responses to environmental change, likely reflecting differences in species assemblages and the effects of other species' traits on the timing and patterns of growth. In plants, decades of research has found consistent relationships between traits and growth strategies under different environments—giving rise to frameworks like the leaf or wood economic spectra[1,2] that separate species along a gradient of fast and acquisitive to slow and conservative strategies. Phenology was foundational to the theory that underlies these spectra[3], but has rarely been included because it is often challenging to measure and highly variable across species when measured in natural conditions.

What hypotheses or questions does this work address? How does budburst phenology fit within plant growth strategies? We test whether early versus late leafout is correlated with acquisitive versus conservative growth strategies, and how these relationships change across spatial scales, sampling from eight sites that span North American.

How does this work advance our current understanding of plant science? We leverage massive experiments with new modeling approaches to show that phenology fits within major frameworks for plant growth, with early to late budburst correlating with plant size and leaf metrics. Further, we found photoperiod appears most linked to other plant traits, despite temperature being the dominant controller of leafout.

Why is this work important and timely? Our results provide the most compelling evidence of trait-phenology relationships across North America with implications for how forest communities will shift with climate change. Further, our novel approach shows how controlled environment experiments combined with cutting-edge hierarchical modeling can elucidate the role of phenology in structuring plant growth strategies.

All authors contributed to this work and approve this version for submission. The manuscript is 4014 words with a 181 word summary, and 4 figures and is not under consideration elsewhere. We hope you find it suitable for publication in *New Phytologist*, and look forward to hearing from you.

Sincerely,
Deirdre Loughnan
Sentinels of Change Postdoctoral Fellow
Hakai Institute | Department of Zoology
University of British Columbia

[1] Wright, I. J., Westoby, M., Reich, P. B., Oleksyn, J., Ackerly, D. D., Baruch, Z., Bongers, F., Cavender-Bares, J., Chapin, T., Cornellissen, J. H. C., Diemer, M., Flexas, J., Gulias, J., Garnier, E., Navas, M. L., Roumet, C., Groom, P. K., Lamont, B. B., Hikosaka, K., Lee, T., Lee, W., Lusk, C., Midgley, J. J., Niinemets, Ü., Osada, H., Poorter, H., Pool, P., Veneklaas, E. J., Prior, L., Pyankov, V. I., Thomas, S. C., Tjoelker, M. G., and Villar, R. The worldwide leaf economics spectrum. *Nature* **428**, 821–827 (2004).

[2] Chave, J., Coomes, D., Jansen, S., Lewis, S. L., Swenson, N. G., and Zanne, A. E. Towards a worldwide wood economics spectrum. *Ecology Letters* **12**(4), 351–366, April (2009).

[3] Grime, J. P. Evidence for the Existence of Three Primary Strategies in Plants and Its Relevance to Ecological and Evolutionary Theory Author (s): J . P . Grime Source : The American Naturalist , Vol . 111 , No . 982 (Nov . - Dec ., 1977), pp . 1169-1194 Published. *The American Naturalist* **111**(982), 1169–1194 (1977).